

Hypergeometric Distribution

Hypergeometric distribution is obtained when the population is finite, the sampling is done without replacement such that events are stochastically independent.

A discrete r.v. X is said to follow a Hypergeometric distribution if it assumes only nonnegative values and its pmf is given as

$$\text{Prob}(X = k) = \frac{{}^M C_k \times {}^{N-M} C_{n-k}}{{}^N C_n}, k=0,1,2,\dots,\min(n,M)$$

here N, M and n are the three parameters of the hypergeometric distribution.

Mean and Variance of Hypergeometric distribution

Mean = $E(X) = \sum_k k \times \text{prob}(X = k) = \frac{nM}{N}$ and variance is

$$\sigma^2 = E(X^2) - \{E(X)\}^2 = \frac{NM(N-M)(N-n)}{N^2(N-1)}$$

Moment generating function of Hypergeometric distribution doesn't exist., but raw moments exist.

Hypergeometric distribution tends to binomial distribution when $N \rightarrow \infty$ and $\frac{M}{N} \rightarrow p$

Examples:

1. A taxi cab company has 12 Ambassadors and 8 Fiats. If 5 of these taxi cabs are in the shop for repairs and Ambassador is as likely to be in for repairs as a Fiat, what is the probability that

- i). 3 of them are Ambassadors and 2 are Fiats
- ii). at least 3 of them are ambassadors?
- iii). all 5 are of them of the same make?

Ans. i) Prob. (3 of them are Ambassadors and 2 are Fiats) = $\frac{{}^{12}C_3 \times {}^8C_2}{{}^{20}C_5}$

ii). Prob.(at least 3 of them are ambassadors) = $\sum_{k=3}^5 \frac{{}^{12}C_k \times {}^8C_{5-k}}{{}^{20}C_5}$

iii). Prob.(all 5 are of them of the same make) = $\frac{{}^{12}C_5 \times {}^8C_0}{{}^{20}C_5} + \frac{{}^{12}C_0 \times {}^8C_5}{{}^{20}C_5}$

2. A lake contains 600 fish, eighty (80) of which have been tagged by scientists. A researcher randomly catches 15 fish from the lake. Find the probability that the number of tagged fish in the researcher's sample are 8?

Ans. Let X is the r.v denotes the number of tagged fishes. X follows Hypergeometric distribution.

$$\text{Prob.}(X=8) = \frac{{}^{80}C_8 \times {}^{600-80}C_{15-8}}{{}^{600}C_{15}}$$

3. A batch of 10 rocker cover gaskets contains 4 defective gaskets. If we draw samples of size 3 without replacement, from the batch of 10, find the probability that a sample contains 2 defective gaskets.

Ans. Let X is the r.v. denotes the number of defective gaskets.

$$\text{Hence, Prob.}(X=2) = \frac{{}^4C_2 \times {}^6C_1}}{{}^{10}C_3}$$

4. Suppose that rockets of a certain types have, by many tests, been established as 90% reliable. Now a modification of the rocket design is being considered. which of the following sets of evidence throws more doubt on the hypothesis that the modified rocket is only 90% reliable:

- i). Of 100 modified rockets tested, 96 performed satisfactorily?
- ii). Of 64 modified rockets tested, 62 performed satisfactorily?